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ECONOMICS · CYBERSECURITY POLICY-ECONOMIC EVIDENCE SYNTHESIS

The Impact of National Cybersecurity Operations Coordination on Critical Economic Sectors and Economic Growth in Nigeria, 2000-2025: A Policy-Economic Evidence Synthesis and Evaluation Framework

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Abstract

Background: Nigeria's economic activity increasingly depends on digital payments, telecommunications, energy control systems, public-sector platforms and other networked services. This dependence increases productivity and market reach but also creates systemic exposure to cyber incidents that can interrupt critical services, destroy data, impose recovery costs and weaken trust. Objective: This study examines how national cybersecurity operations coordination can protect critical economic sectors and support economic resilience in Nigeria over 2000-2025, while distinguishing defensible evidence from causal claims that the available national time series cannot sustain. Methods: A longitudinal policy-economic evidence synthesis was conducted using Nigeria's cybersecurity laws and strategies, the National Cybersecurity Coordination Centre and ngCERT operational model, the 2024 Critical National Information Infrastructure Order, Central Bank of Nigeria cybersecurity frameworks, International Telecommunication Union capability benchmarks, Nigerian Communications Commission evidence on cybercrime and digital-sector exposure, and World Bank macroeconomic indicators. The study maps institutional milestones, identifies economic transmission channels, and specifies a sector-panel and interrupted-time-series evaluation design for future causal testing. Results: Nigeria's architecture evolved from fragmented sector controls toward a coordinated national model centred on NCCC, ngCERT and sector CSIRTs. The 2024 CNII Order designates thirteen economic sectors for critical protection. ITU's 2024 Global Cybersecurity Index classifies Nigeria in Tier 3, with strongest performance in legal and technical measures and weaker capacity-development and cooperation scores. Earlier Nigerian evidence estimated cybercrime losses at about 0.08% of GDP in 2016, while current ngCERT advisories show persistent phishing, ransomware and exploitation risks to finance, telecommunications, government and other critical services. Conclusion: National cyber operations coordination is best interpreted as resilience infrastructure: its economic value arises from avoided loss, shorter disruption, faster recovery, reduced cross-sector contagion and stronger confidence in digital services. A credible growth estimate requires harmonized incident, downtime, response-time, loss and sector-output data; the paper therefore provides a reproducible evaluation framework rather than an invented treatment effect.

Keywords: cybersecurity economics; National Cybersecurity Coordination Centre; ngCERT; critical national information infrastructure; economic growth; operational resilience; Nigeria

1. Introduction

Nigeria's economy has become more digitally mediated across banking, telecommunications, government services, identity systems, transport, health, energy and commerce. Digitalisation lowers transaction costs and extends service reach, but it also converts cyber availability, integrity and trust into economic inputs. A successful ransomware campaign against a payment processor, a compromise of telecommunications infrastructure, or a disruptive attack on operational technology can therefore move quickly from a technical event to lost transactions, service downtime, recovery expenditure, reputational damage and reduced confidence. The economic problem is not simply the number of attacks; it is the extent to which incidents disrupt productive activity and propagate across interdependent sectors.

Cyber risk has the characteristics of an externality and a coordination problem. Organisations may underinvest when the benefits of their security spending spill over to customers, suppliers and other networks, while attackers exploit the weakest connected entity. The economic literature accordingly treats cybersecurity as a domain of incomplete information, misaligned incentives and interdependence. Recent work also shows that cyber incidents can depress productivity and growth by forcing labour and capital away from productive use and into recovery. A national coordination function can address part of this market failure by sharing threat intelligence, coordinating incident response, setting common protection baselines and reducing the time between local detection and cross-sector action (Abrardi, Comino, & Grassini, 2026; Uguet & Jung, 2026).

Nigeria's institutional architecture has strengthened in stages. The Cybercrimes (Prohibition, Prevention, etc.) Act 2015 created a statutory basis for national coordination. The National Cybersecurity Policy and Strategy 2021 places the National Cybersecurity Coordination Centre (NCCC) at the centre of implementation and coordination. The Nigeria Computer Emergency Response Team (ngCERT) operates as the national incident-response capability under the NCCC. In 2024, the Cybercrimes Act was amended, the Central Bank of Nigeria issued a revised risk-based cybersecurity framework for deposit money banks and payment service banks, and the Designation and Protection of Critical National Information Infrastructure Order formally identified thirteen critical sectors. These instruments transform cybersecurity from a narrow IT-control problem into an issue of economic continuity and national resilience.

The source study proposed an examination of a national cybersecurity operations centre over 2000-2025. For institutional accuracy, this paper uses Nigeria's official terminology - NCCC, ngCERT and the sector CSIRT network - while retaining the underlying research question: whether coordinated national cyber operations can protect critical economic activity and contribute to growth. The paper does not assign a causal coefficient to NCCC because a consistent annual national series on incidents, mean time to detect, mean time to recover, sector downtime and avoided loss is not publicly available for the entire period. Instead, it combines policy history, economic evidence and operational metrics into a transparent framework that can support a later causal evaluation.

2. Cybersecurity, Critical Sectors and the Nigerian Economy

2.1 Digital dependence and economic exposure

The economic importance of cybersecurity rises with digital dependence. World Bank indicators show that Nigeria's GDP reached about US\$290.8 billion in 2025 and real GDP growth was approximately 4.0%, while internet use had expanded substantially by the mid-2020s. The Nigerian Communications Commission's sector statistics also show a large telecommunications contribution to GDP over the last decade, although recent figures require caution because the National Bureau of Statistics rebased GDP. These indicators do not measure cybersecurity, but they show why service availability and confidence in digital infrastructure matter to aggregate output.

An older NCC-commissioned study on cybercrime, foreign direct investment and national development reported that the Federal Government estimated the annual cost of cybercrime in 2016 at about 0.08% of GDP, roughly N127 billion at the time. Such figures are necessarily approximate because many incidents are unreported and indirect losses are difficult to observe. They nonetheless illustrate a central economic point: the loss function extends beyond stolen funds to business interruption, remediation costs, intellectual-property loss, reputational damage, weakened investor confidence and reduced trust in online services.

2.2 Critical National Information Infrastructure

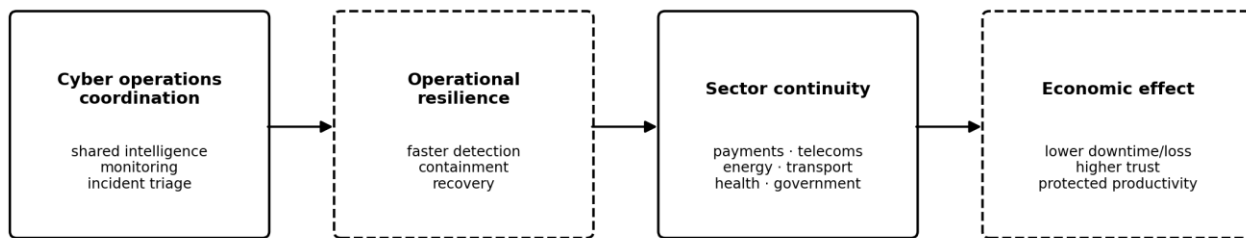
The 2024 Designation and Protection of Critical National Information Infrastructure Order links cybersecurity directly to national imperatives, economic development, public health, safety, defence and government operations. It designates information and communications systems in thirteen economic sectors as Critical National Information Infrastructure (CNII). The Order also requires protection planning, information sharing, audits and implementation

measures. This creates a legal basis for treating cyber resilience as a system-wide economic policy issue rather than a collection of isolated organisational security programmes.

No.	CNII sector	Economic / operational significance
1	Power and energy	Grid, generation, oil and gas, energy-management and control systems
2	Water	Dams, water treatment, distribution and monitoring systems
3	Information, communication, science and technology	Telecommunications, internet, digital identity and backbone services
4	Banking, finance and insurance	Payment, settlement, banking, insurance and financial-market infrastructure
5	Health	Hospitals, health systems, public-health and health-data services
6	Public administration	Core government platforms, databases and service-delivery systems
7	Education	Institutional networks, student systems and digital learning infrastructure
8	Defence and security	Defence, security and intelligence information systems
9	Transport	Aviation, maritime, road, rail and associated information systems
10	Food and agriculture	Food systems, agricultural information and production-support infrastructure
11	Safety and emergency services	Emergency coordination, response and public-safety systems
12	Industrial and manufacturing	Industrial control, manufacturing and supply-chain systems
13	Mines and steel	Mining, steel and associated operational-information systems

Table 1. Critical sectors designated by Nigeria's 2024 CNII Order; descriptions summarize economic functions rather than reproduce the Schedule verbatim.

Cyber-resilience transmission pathway to economic performance



The growth channel is protective and enabling: coordination can avert losses and preserve digital output; it does not mechanically create GDP.

Figure 1. Cyber-resilience transmission pathway from national coordination to economic continuity.

3. Theoretical Framework: Cyber Resilience as Economic Infrastructure

3.1 Expected-loss economics

Cybersecurity expenditure is economically justified when the expected value of avoided losses and continuity gains exceeds the cost of controls. At sector s and time t , expected cyber loss can be represented as the probability of a material incident multiplied by the combined direct, downtime and spillover consequences:

$$ECL(s,t) = P(Incident(s,t)) \times [DirectLoss(s,t) + DowntimeLoss(s,t) + SpilloverLoss(s,t)]$$

National coordination can affect both terms. Threat intelligence, common reporting and exercises can reduce the probability that known techniques succeed; coordinated containment, forensics and recovery can reduce loss severity after compromise. The net resilience dividend is therefore the expected baseline loss minus residual loss after coordination and minus the resource cost of operating the national and sector response architecture.

$$Resilience\ Dividend = ECL(baseline) - ECL(coordinated) - Coordination\ Cost$$

3.2 Public-good and network-externality logic

Cybersecurity is interdependent. An institution that rapidly reports a ransomware indicator may enable dozens of other operators to block the same campaign before compromise. Conversely, an unprotected supplier can transmit risk into multiple critical operators. Information sharing, common minimum controls and sector CSIRTs therefore produce spillovers that private firms may not fully internalise. This public-good logic supports a national coordination layer, provided that it does not displace the responsibility of operators to maintain their own governance, security operations, backups and incident-response capacity.

3.3 Growth and productivity channel

A cybersecurity coordination institution should not be modelled as a direct production factor in the same way as labour or physical capital. Its economic function is closer to risk-adjusted infrastructure. By reducing expected downtime, loss and uncertainty, resilience can protect the effective utilisation of digital capital, support transaction continuity and preserve confidence. A growth equation therefore requires cyber-resilience variables alongside digital intensity and standard macroeconomic controls:

$$\Delta \ln(Y_t) = \alpha + \beta \cdot \text{CyberResilience}(t) + \gamma \cdot \text{DigitalIntensity}(t) + \delta X(t) + \varepsilon(t)$$

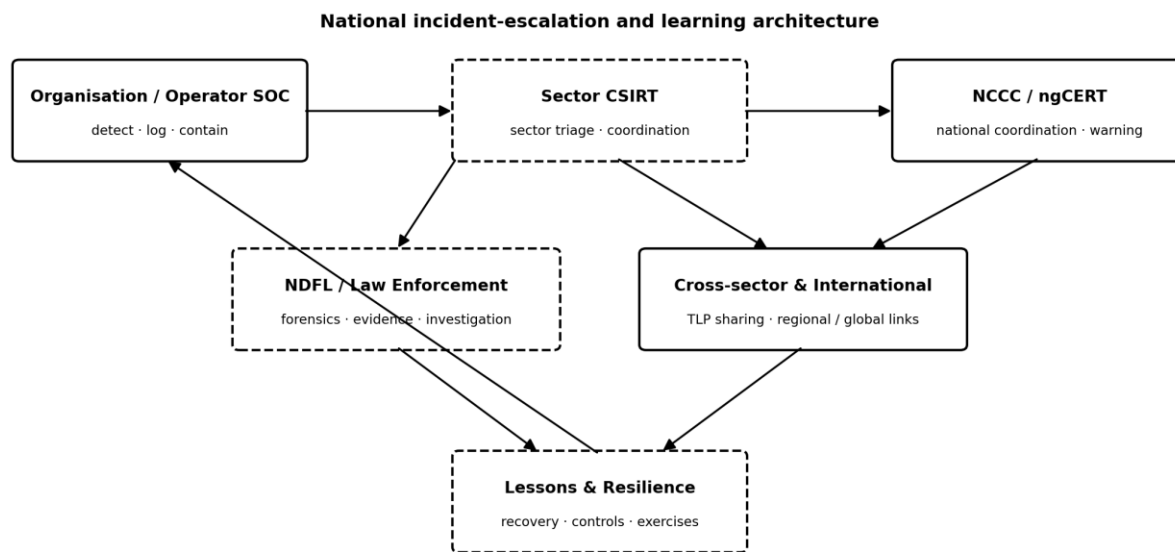
Here Y is real output, DigitalIntensity captures the economy's dependence on ICT-enabled activity, and X includes capital formation, labour, inflation, exchange-rate conditions, trade exposure and major shocks. The coefficient β should be interpreted causally only when the resilience measure has a credible source of exogenous variation or a well-specified counterfactual.

4. Institutional Evolution, 2000-2025

Period	Milestone	Economic-resilience significance
2000-2013	Digital expansion with fragmented sector security controls	Increasing economic exposure without a single mature national coordination architecture.
2014	Maiden national cybersecurity policy/strategy era	Formal national strategic direction emerges.
2015	Cybercrimes (Prohibition, Prevention, etc.) Act	Statutory basis for national cybersecurity coordination and protection of critical systems.
2018	CBN risk-based cybersecurity framework for banks/payment providers	Sector-specific minimum controls and supervisory expectations for finance.
2021	National Cybersecurity Policy and Strategy 2021	NCCC entrusted with harmonising national cybersecurity efforts; strategy links cyber security to national prosperity.
2023	Nigeria Data Protection Act	Stronger legal framework for personal-data protection and breach accountability.
2024	Cybercrimes Amendment Act; revised CBN framework; CNII Order	Stronger statutory and sector controls; thirteen sectors designated as CNII; resilience obligations broaden.
2025	NIST SP 800-61 Rev. 3 and continuing Nigerian operationalisation	International incident-response practice shifts toward integrated risk-management and continuous learning; useful benchmark for sector CSIRTs.

Table 2. Key institutional milestones used for periodisation. The table supports policy analysis; it is not a causal event-study result.

The 2021 strategy is especially important because it treats national cybersecurity as a coordinated programme rather than a collection of independent agency actions. NCCC's current mandate covers national coordination, protection of CNII, threat-intelligence sharing, incident-response coordination, capacity building and international cooperation. ngCERT provides national incident response and serves government, critical infrastructure operators, sector CSIRTs, public and private organisations. This arrangement is economically relevant because cross-sector incidents cannot be efficiently managed when every operator acts in isolation.



Architecture reflects the official NCCC/CSIRT coordination model; organisational security teams remain responsible for local detection and first response.

Figure 2. National incident-escalation and learning architecture adapted to Nigeria's NCCC / ngCERT / sector-CSIRT model.

5. Method and Evidence Design

5.1 Research design

The study uses a longitudinal policy-economic evidence-synthesis design covering 2000-2025. It is retrospective and observational. It combines documentary analysis of laws, strategies, operational mandates and regulatory frameworks with descriptive macro-digital indicators and an explicit econometric design for future testing. This structure is chosen because a national operations centre cannot be credibly evaluated using a simple before-after GDP comparison: Nigeria's growth over the period was simultaneously affected by oil-price cycles, exchange-rate regimes, population growth, recessions, the COVID-19 shock, inflation, insecurity, telecommunications expansion and multiple financial-sector reforms.

5.2 Evidence sources

Evidence domain	Principal sources	Use in study
National cyber governance	Cybercrimes Act 2015 and 2024 amendment; NCPS 2021; CNII Order 2024; NCCC and ngCERT mandates	Institutional authority, critical-sector coverage, operational roles
Financial-sector resilience	CBN risk-based cybersecurity frameworks, 2018 and 2024	Sector controls, governance, incident response, metrics and supervision
National capability benchmark	ITU Global Cybersecurity Index 2024	Legal, technical, organisational, capacity and cooperation scores
Cyber-risk exposure	NCC cybercrime/FDI study; ngCERT advisories	Loss estimates, prevalent threats, sector exposure
Macro-digital economy	World Bank WDI; NCC sector statistics	GDP, growth, internet use, telecommunications contribution and digital dependence
International technical benchmark	NIST CSF 2.0; NIST SP 800-61 Rev. 3	Governance and incident-response outcome structure

Table 3. Evidence architecture and analytical use.

5.3 Measurement strategy

A defensible national evaluation should separate capability, incident performance, economic loss and macroeconomic outcomes. Capability indicators measure whether institutions and controls exist; operational indicators measure whether incidents are detected and resolved more effectively; economic indicators measure avoided disruption and loss; macro outcomes capture whether these gains are large enough to affect sector output or aggregate growth. Collapsing these levels into a single 'cybersecurity score' risks spurious inference.

Layer	Candidate indicators	Preferred frequency
Capability	Sector CSIRT coverage; regulatory compliance; exercise frequency; asset visibility	Annual / quarterly
Detection	Mean/median time to detect; % incidents detected internally; log coverage	Incident / monthly
Response	Mean time to contain; escalation latency; severity-normalised response time	Incident
Recovery	Mean time to recover; backup restore success; service-hours lost	Incident
Loss	Direct loss; remediation cost; business-interruption value; insurance recovery	Incident / annual
Spillover	Number of affected organisations/sectors; cross-sector propagation	Incident
Trust	Digital transaction continuity; customer complaints; service abandonment; investment response	Monthly / quarterly
Economic output	Sector value added; real GDP growth; productivity; investment	Quarterly / annual

Table 4. Measurement hierarchy for a national cyber-operations economic evaluation.

5.4 Proposed econometric identification

The preferred empirical design is a sector-by-time panel rather than a single national annual series. Sectors differ in digital intensity, exposure and timing of regulatory implementation, creating potentially useful variation. A fixed-effects specification can compare changes within sectors while controlling for common macro shocks:

$$Growth(s,t) = \alpha s + \lambda t + \beta \cdot CyberOpsIntensity(s,t) + \theta Z(s,t) + \varepsilon(s,t)$$

CyberOpsIntensity should be built from observed operational metrics rather than policy-announcement dummies alone. Where exact implementation dates are available, interrupted-time-series or staggered difference-in-differences designs may be considered, but only after testing pre-trends and accounting for contemporaneous reforms. The 2024 CNII designation is not, by itself, an exogenous treatment because sectors were selected precisely because they were critical. Any evaluation must therefore avoid interpreting designation as random assignment.

5.5 Validity and evidence discipline

The study uses three safeguards. First, it distinguishes legal or institutional commitment from operational performance. Second, it does not infer economic causality from the temporal coincidence of stronger cyber policy and GDP growth. Third, it treats public loss estimates as lower-bound or approximate measures because incident under-reporting, confidential settlements and intangible losses are substantial. These constraints are central to the paper's contribution: a useful economic model must make uncertainty explicit rather than compensate for missing data by inventing precision.

6. Findings from the Evidence Synthesis

6.1 Nigeria has moved toward a coordinated national cyber-operations model

The strongest documentary finding is institutional consolidation. The NCCC is now the national coordinating authority under the Office of the National Security Adviser, while ngCERT provides national incident-response capability and sector CSIRTs form the operational bridge between individual operators and national coordination. The escalation logic published by NCCC is clear: incidents begin at the organisation, move to the sector CSIRT where wider sector impact is assessed, and reach NCCC/ngCERT when they cross sectors or become nationally significant. This reduces ambiguity over responsibility and creates a mechanism for turning one organisation's incident into early warning for others.

6.2 Critical-infrastructure protection is now economically explicit

The 2024 CNII Order materially strengthens the economic framing of cybersecurity. Its objectives explicitly refer to continued operation and to systems that drive economic development, public health, safety, defence and government operations. By naming thirteen sectors, the Order recognises cyber interdependence beyond banking and telecommunications. This matters because national loss from a cyber incident depends on network effects: an energy failure can impair telecommunications; telecommunications disruption can impair payments; payment disruption can propagate into retail, logistics and public services.

6.3 National capability is substantial but uneven

ITU's Global Cybersecurity Index 2024 places Nigeria in Tier 3, 'Establishing'. Nigeria scored 19.52/20 on legal measures and 18.88/20 on technical measures, compared with 16.57 for organisational measures, 13.15 for capacity development and 14.28 for cooperation. The pattern is analytically useful: law and technical structures are relatively strong, while workforce, institutional integration and cooperation remain areas where marginal investment may generate high resilience returns. The index measures commitment and capability, not the number of prevented attacks or the GDP effect of NCCC.

GCI 2024 pillar	Nigeria score	Interpretation
Legal measures	19.52	Relative strength
Technical measures	18.88	Relative strength
Organisational measures	16.57	Established but below legal/technical
Capacity development	13.15	Priority growth area
Cooperation measures	14.28	Priority growth area
Overall tier	Tier 3	Establishing

Table 5. Nigeria's Global Cybersecurity Index 2024 profile (ITU, 2024). Scores are commitment/capability measures, not economic-impact estimates.

6.4 Threat pressure remains material

ngCERT advisories in 2024-2026 report ransomware, phishing, credential harvesting, exploitation of unpatched systems and attacks against cloud and telecommunications services used by government, finance, education, health and other critical organisations. This persistence means that stronger national capacity should not be evaluated by a fall in raw incident counts alone. Better monitoring can initially increase reported incidents. More meaningful outcomes are severity-adjusted losses, time to detection, time to containment, time to recovery, cross-sector propagation and the proportion of incidents detected before material business disruption.

6.5 Economic contribution is primarily protective

The evidence supports a bounded economic claim. Cyber operations coordination can protect growth by reducing the expected cost of disruption and by preserving trust in digital services; it cannot, by itself, guarantee higher GDP growth. The mechanism is strongest in sectors with high digital intensity and systemic connectivity. The financial system illustrates this logic: the CBN's 2024 framework emphasises governance, risk management, resilience, emerging

technologies, metrics, monitoring and compliance because cyber failure can impose financial loss and undermine public confidence. Similar logic applies to power, telecommunications, health and government platforms.

7. National Cyber-Operations Economic Resilience Scorecard

For policy and future research, the most useful next step is a national scorecard that links technical performance to economic outcomes. It should be maintained at sector level and aggregated only after standardising definitions. The scorecard below is deliberately outcome-oriented and can be implemented without exposing sensitive incident details.

Dimension	Metric	Frequency	Economic interpretation
Coverage	% designated CNII operators connected to sector CSIRT / national reporting	Quarterly	Measures reach of coordination
Detection	Median time to detect high-severity incidents	Monthly	Lower is better; stratify by sector and severity
Escalation	Median time from operator detection to sector/national notification	Incident	Measures coordination friction
Containment	Median time to contain high-severity incidents	Incident	Direct operational resilience
Recovery	Service restoration time and backup-recovery success	Incident	Links controls to continuity
Propagation	Incidents affecting >1 organisation or sector	Quarterly	Measures systemic-risk containment
Economic loss	Direct + interruption loss per severity-adjusted incident	Quarterly	Primary economic damage measure
Averted loss	Counterfactual loss avoided through early warning / shared indicators	Case-based	Requires documented counterfactual
Exercises	Critical-sector exercises completed and findings closed	Annual	Measures preparedness and learning
Workforce	Certified/competent analysts per critical operator and CSIRT	Annual	Addresses capability bottleneck
Trust/continuity	Payment/telecom/service availability during incidents	Monthly	Connects cyber resilience to user-facing economy

Table 6. Proposed National Cyber-Operations Economic Resilience Scorecard.

The scorecard should be accompanied by a severity taxonomy and confidentiality controls. Public reporting can aggregate indicators while sensitive technical details remain protected under appropriate information-sharing protocols. NCCC/ngCERT's use of Traffic Light Protocol concepts is relevant: economic evaluation does not require publication of exploitable details, only standardised outcome measures.

8. Policy Implications

First, Nigeria should treat cyber resilience as productive infrastructure. Budget appraisal for critical-sector cybersecurity should compare control costs with expected avoided loss, downtime and systemic spillover, rather than treating cybersecurity as an overhead detached from service performance. Major digital infrastructure projects should include measurable security, recovery and incident-reporting outcomes from design through operation.

Second, priority should shift from institution-building alone to operational performance. The existence of laws, centres and frameworks is necessary but not sufficient. NCCC and sector regulators should standardise a small set of incident-performance metrics, including severity-normalised detection, containment and recovery times. A national dashboard can report trends without disclosing sensitive technical details.

Third, sector CSIRTs should be strengthened as the main transmission layer between organisation-level security operations and national coordination. Sector teams understand specialised technologies and can distinguish routine incidents from systemic events. Regular exercises should test cross-sector dependencies such as power-telecommunications-payments and identity-government-service chains.

Fourth, capacity development and cooperation deserve particular emphasis. Nigeria's ITU profile shows these pillars lagging legal and technical measures. Investment in analysts, digital forensics, threat-intelligence practice, incident command, industrial-control-system security and executive cyber-risk governance can therefore complement existing legal authority. Universities and professional bodies should be integrated into the training pipeline.

Fifth, economic research should be embedded into incident-response governance. A standard post-incident economic-impact form should capture direct financial loss, service-hours lost, staff recovery time, customer impact, third-party propagation and restoration cost. These data would allow economists to move from broad national loss claims to sector-specific estimates of resilience dividends.

9. Limitations and Research Priorities

This paper does not estimate a causal GDP effect for NCCC or ngCERT. The national coordination architecture emerged gradually, its intensity is not observed consistently from 2000, and macroeconomic outcomes are affected by numerous confounders. Public incident data are incomplete and reporting improves as capability matures, creating measurement bias. Telecommunications contribution to GDP is also affected by national-account rebasing, so longitudinal comparisons require harmonised NBS series.

Future research should construct a secure sector panel using regulator and operator records. The strongest design would combine quarterly sector output with incident severity, MTTD, MTTR, downtime, cybersecurity spending, digital intensity and implementation dates of sector controls. Synthetic-control, interrupted-time-series or difference-in-differences methods could then be considered where assumptions are defensible. Case studies of major incidents should complement econometrics because cross-sector propagation and recovery decisions are not fully captured by aggregate data.

10. Conclusion

Nigeria's cybersecurity architecture has evolved from dispersed sector controls toward a nationally coordinated model centred on NCCC, ngCERT and sector CSIRTs. The 2024 CNII Order makes the economic stakes explicit by designating thirteen sectors whose digital systems are critical to national development, public safety and government operations. Evidence from national reports, sector regulation, ITU capability benchmarks and current threat advisories supports the proposition that cyber operations coordination can protect economic activity by reducing disruption, loss and systemic contagion.

The appropriate economic claim is therefore a resilience claim, not an automatic growth claim. A national coordination centre creates value when it helps operators detect earlier, contain faster, recover more reliably and warn others before attacks spread. Measuring that value requires incident-level and sector-level operational data linked to economic outcomes. The evaluation framework and scorecard developed in this article provide a route from policy commitment to measurable economic evidence without inventing a causal effect that the current public data cannot support.

Declarations

Ethics and participant involvement

Not applicable. This policy-economic study uses publicly available documentary, regulatory, statistical and scholarly sources and involves no human participants, animals, patient material or identifiable personal records.

Data availability

No confidential incident dataset was used. The public evidence sources are identified in the reference list. World Bank, ITU, NCCC, ngCERT, CBN and NCC data and documents remain subject to the access conditions and update practices of their respective publishers.

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The author declares no competing interest that would alter the interpretation of the policy-economic analysis presented in this article.

Author contribution and accountability

Dr. Ibora Tom IDIM is the sole named author of e060 and accepts responsibility for the economic framing, evidence synthesis, interpretation and final manuscript. Editorial and document-production assistance is not treated as authorship or as a source of evidence.

Figure and table provenance

Figures 1-2 and Tables 1-6 were created specifically for e060 from the cited public evidence and analytical framework. They contain no participant-level data and reproduce no third-party copyrighted image.

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